Week 3: Linux Logs, Sysmon, auditd and Snort Setup

Theory Notes, Developing Basic Understanding in SOC

Question 3. What is auditd on Linux? How does it work, and what is the difference between auditd and syslog? Include a table comparing the two.

What is auditd

auditd is the part of the Linux Audit Framework that runs in user space. The audit system itself is built into the Linux kernel. Together they record security related activity on the machine, such as commands being run, files being opened or changed, and users logging in. The records are saved in `/var/log/audit/audit.log`, and many organisations use them for security monitoring and compliance.

How it works

- The kernel notices an event, for example when a program is executed through the `execve` system call.
- Audit rules decide which events should be recorded. These can be added with `auditctl` or saved in `/etc/audit/rules.d/`.
- The kernel passes the matching events to the auditd service, which writes them into audit.log.
- To read the logs, I can use `ausearch` to search for events and `aureport` to see summaries.
- A rule can be given a key with `-k`, which makes searching easier, for example `ausearch -k nano`.

Fields in an audit log entry

Field	What it tells me
type	The kind of record, such as EXECVE, SYSCALL, USER_LOGIN, USER_AUTH or PATH
msg=audit(time:serial)	The time of the event and a number that groups records belonging to the same event
uid and auid	The user ID, and the original login user ID which stays the same even after using sudo
exe and comm	The full path and the short name of the program
a0, a1 and so on	The arguments of the command that was run
key	A label I give to a rule so I can find its events easily

auditd compared with syslog

Point	auditd	syslog
Where it comes from	The kernel audit system	Applications, services and the system
Purpose	Security auditing and tracking exact actions	General system and application messages
Level of detail	Very detailed, including system calls, users and arguments	Short messages, and the detail depends on the application
Log file	/var/log/audit/audit.log	/var/log/syslog or /var/log/messages

Point	auditd	syslog
Format	key=value records, often several lines for one event	One line per message with time, host and text
Configuration	Audit rules, such as watching a file or a system call	Rules based on facility and severity
Protection	Rules can be locked until the next reboot	Easier to edit or fill with fake messages
Use in a SOC	Tracking commands, file changes and use of privileges	Checking service status and general errors

In simple words, syslog tells me that something happened in a service, while auditd tells me which user ran which command on which file.